11.

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Maithreyan.P *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	

Total Marks 31 /40

Signature of the Faculty

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Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

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of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

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Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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1. A company needs to transfer a 150MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The transport layer divides the file into 1500-byte segments, while the data link layer encapsulates each segments into 1500 by adding 40 bytes of header and trailer. calculate the total number of segments and frams transmitted, the total transmission overheaded Explain how the transport and data link layers ensure reliable communication.

2. A transport layer protocol used a sliding window size of 6 frames. Each fram carries 1024 bytes of user data. If the sender transmits 48 frames, calculate number of transmitted windows required. If one, frame in a window is lost and munst retransmitted Explain how flow control improves communication efficiency.

3. A 12,000-byte IP packet transmitted through a router where the Maximum transmission unit (MTU) is 1500 bytes. Assuming each fragment requires a 20 byte IP header, Explain how the Network layer ensures successful delivery of fragmented packets the destination.

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A. A transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1000 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission window required. If one frame in every window is lost and must be retransmitted Explain how flow control improves communication efficiency.

CO1 AT-1

Short Answers

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1. The Given:

file size : 150MB

Band width : 50Mbps

segment size : 1500 bytes

frame overheaded = 40 bytes.

* To calculate a total segments :

$$= \frac{150 \times 10^6}{1500}$$

= 100,000 segments.

* To find total frames that should be a segment,

Total Fame = 100,000 frames

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overheader = 100,000 x 40

Total overhead = 4,000,000 bytes

= 4MB

finally, Total data transmitted is

= 150 + 4 = 154MB.

Let us calculated the transmission time,

= 154 x 8 / 50

$= \frac{1232}{50}$

= 24.64 seconds.

Explaination:

The transpart layer divides the file into segments and uses acknowledgements and retransmission to ensure reliable delivery The data link layer converts segments into frames and uses error detection to identify damaged frames.

sliding window and flow control,

Given:

Window size: 6 frames

Data per frame: 1024 bytes.

Total frames: 48

calculation:

No. of windows = $\frac{48}{6}$ = 8 windows

one last frame in each window = 1

retransmission.

Total retransmission = 8 × 1 = 8.

Explaination:

Flow control prevents the sender from sending data faster than the receiver can handle.

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The sliding window allows multiple frames to be sent before receiving the acknowledgements improving network efficiency.

Answer: 8 windows and 8 retransmission.

The session layer creats synchronization checkpoints during data transfer. If a failer will be done in the file transfer. the transmission and restart from the last checkpoints are insted of starting from the begining.

3. Given:

IP packet size = 12,000 bytes

MTU = 1500 bytes

Header size per fragment = 20 bytes.

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1. Number of fragments =

$$\Rightarrow \frac{12000}{1500} = 8 \text{ fragments}$$

To calculate the total header overhead is upto 20 bytes.

Total overhead = $8 \times 20 = 160$ bytes.

Explanation:

IP Packet fragmentation:

The Network layer divides a large IP packet into smaller fragments according to the MTU.

Each fragment contains a header with identification and fragment information.

The destination uses this information to reassemble the original packet.

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Answer :

IP packet fragmentation:

8 fragments and 160 bytes of the header and overhead.

working :

Flow control manages the speed of data transmission between sender and receiver. The sliding window allows several frames to be sent together reducing waiting time and improving the data.

40 Given:

Total file size : 600 MB

checkpoint intervals = 60 MB

Failure occurs after 390 MB

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calculations:

1. Number of checkpoints:

$$= \frac{360}{60} = 6,$$

completed checkpoints are 60, 120, 180, 240, 300 and 360 MB.

Therefore 6 checkpoints were created before failure.

2. Data to retransmit = 390 − 360 = 30 MB

Answer:

6 checkpoints and 30MB to retransmit

Explaination:

The session layer creates synchronization checkpoints during data transfer. If a failure occurs, transmission can restart from the last checkpoint instead of start

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